KAIZEN MATHS

Lesson starter

Hyperbolic Functions

Hyperbolic Calculus

Name _____

S1

Differentiate each function with respect to x.

Differentiate $y = \cosh(2x + 1)$.

S2

Integrate each function with respect to x.

Find $\int \tanh(5x)\, dx$.

S3

Differentiate each product using the product rule.

Differentiate $y = x \cosh(4x)$.

S4

Find the second derivative.

For $y = \cosh(4x)$, find $\frac{d^2y}{dx^2}$.

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ALG-6FDD-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Hyperbolic Functions

Hyperbolic Calculus

Name _____

E1

Differentiate each function with respect to x.

Differentiate $y = \cosh(3x - 1)$.

E2

Integrate each function with respect to x.

Find $\int \cosh(4x)\,dx$.

E3

Differentiate each product using the product rule.

Differentiate $y = x \sinh(2x)$.

KAIZEN MATHS

Exit ticket

Hyperbolic Functions

Hyperbolic Calculus

Name _____

E1

Differentiate each function with respect to x.

Differentiate y = cosh(3x - 1).

E2

Integrate each function with respect to x.

Find ∫ cosh(4x) dx.

E3

Differentiate each product using the product rule.

Differentiate y = x sinh(2x).

ALG-6FDD-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Hyperbolic Functions

Hyperbolic Calculus

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Differentiate each function with respect to x.

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Find ∫ cosh(4x) dx.

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Differentiate each product using the product rule.

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ALG-6FDD-L3 · E

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Hyperbolic Calculus

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Differentiate each function with respect to x.

Differentiate y = cosh(3x - 1).

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Integrate each function with respect to x.

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Differentiate each product using the product rule.

Differentiate y = x sinh(2x).

ALG-6FDD-L3 · E

Show your working

Teacher answers & working

Hyperbolic Functions · Hyperbolic Calculus

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S1

Differentiate each function with respect to x.

Differentiate y = cosh(2x + 1).

Answer:

$\frac{dy}{dx} = 2\sinh(2x + 1)$

Working:

Let u = 2x + 1. Then $\frac{du}{dx}$ = 2.

The derivative of cosh u is sinh u.

Use the chain rule.

$\frac{dy}{dx} = 2\sinh(2x + 1)$.

S2

Integrate each function with respect to x.

Find ∫ tanh(5x) dx.

Answer:

$\frac{1}{5}\ln(\cosh(5x)) + C$

Working:

Let u = 5x. Then $\frac{du}{dx}$ = 5.

So $dx = \frac{du}{5}$.

∫ tanh(5x) dx = $\frac{1}{5}\int \tanh u \, du$.

The integral of tanh u is ln(cosh u).

Therefore the integral is $\frac{1}{5}\ln(\cosh(5x)) + C$.

S3

Differentiate each product using the product rule.

Differentiate y = x cosh(4x).

Answer:

$\frac{dy}{dx}$ = cosh(4x) + 4x sinh(4x)

Working:

Use the product rule: $\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$.

Let u = x and v = cosh(4x).

$\frac{du}{dx} = 1$.

$\frac{dv}{dx} = 4\sinh(4x)$.

$\frac{dy}{dx}$ = x(4sinh(4x)) + cosh(4x)(1).

Therefore $\frac{dy}{dx}$ = cosh(4x) + 4x sinh(4x).

Teacher answers & working

Hyperbolic Functions · Hyperbolic Calculus

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S4

Find the second derivative.

For y = cosh(4x), find $d^2y\ dx^2$.

Answer:

$\frac{d^2y}{dx^2} = 16\cosh(4x)$

Working:

Differentiate once.

$\frac{dy}{dx} = 4\sinh(4x)$.

Differentiate again, remembering the chain rule.

$\frac{d^2y}{dx^2} = 16\cosh(4x)$.

E1

Differentiate each function with respect to x.

Differentiate y = cosh(3x - 1).

Answer:

$\frac{dy}{dx} = 3\sinh(3x - 1)$

Working:

Let u = 3x - 1. Then $\frac{du}{dx}$ = 3.

The derivative of cosh u is sinh u.

Use the chain rule.

$\frac{dy}{dx} = 3\sinh(3x - 1)$.

E2

Integrate each function with respect to x.

Find ∫ cosh(4x) dx.

Answer:

$\frac{1}{4}\sinh(4x) + C$

Working:

Let u = 4x. Then $\frac{du}{dx}$ = 4.

So $dx = \frac{du}{4}$.

∫ cosh(4x) dx = $\frac{1}{4}\int \cosh u\ du$.

The integral of cosh u is sinh u.

Therefore the integral is $\frac{1}{4}\sinh(4x) + C$.

Teacher answers & working

Hyperbolic Functions · Hyperbolic Calculus

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E3

Differentiate each product using the product rule.

Differentiate y = x sinh(2x).

Answer:

$\frac{dy}{dx}$ = sinh(2x) + 2x cosh(2x)

Working:

Use the product rule: $\frac{d}{dx}$(uv) = u$\frac{dv}{dx}$ + v$\frac{du}{dx}$.

Let u = x and v = sinh(2x).

$\frac{du}{dx} = 1$.

$\frac{dv}{dx} = 2\cosh(2x)$.

$\frac{dy}{dx}$ = x(2cosh(2x)) + sinh(2x)(1).

Therefore $\frac{dy}{dx}$ = sinh(2x) + 2x cosh(2x).